KAZAKH-BRITISH TECHNICAL UNIVERSITY

JSC «Kazakh British Technical University»
School of IT and Engineering

APPROVED BY
Dean of SITE
Azamat Imanbayev

«05» 01 **2026**

SYLLABUS

Discipline: INFT4245 Golang application development
Term: Spring 2026
Instructors' full name: Salikh Pernebek, Bakytzhan Nurmagambetov

Personal Information about the Instructor	Time and place of classes		Contact information	
	Lessons	Office Hours	Tel.	e-mail
Salikh Pernebek Lecturer	According to the schedule	According to the schedule	+7 776 996 1414	s.pernebek@kbtu.kz
Bakytzhan Nurmagambetov Tutor	According to the schedule	According to the schedule		

COURSE DURATION: 3 hours a week, 15 weeks, 45 class hours

MS Teams code
General | Golang Application Development | Microsoft Teams

GENERAL COURSE AIMS:

1. Learn the basics of idiomatic Go for backend services, including HTTP, data storage, testing, and deployment.
2. Learn concurrency, context, timeouts, graceful shutdown, and reliability patterns for services.
3. Deliver a production-like web API with auth, storage, tests, and containerization.

COURSE DESCRIPTION

This is a practical course covering the fundamentals of Go programming and backend development. Students learn routing, data storage, testing, concurrency, and basic deployment to build fast, secure, and maintainable REST services. The course blends theoretical lectures with hands-on labs and a team project to reflect real industry practice.

COURSE OBJECTIVES

- To introduce students to Go programming basics and idiomatic style.
- To teach web service development with HTTP, routing, JSON, and middleware.
- To provide hands-on practice with database, migrations, testing, and concurrency.
- To guide students through a real project with team collaboration and basic deployment.

COURSE OUTCOMES

1. Students will learn the basics of backend development using Go.
2. Students will understand service architecture by building a working web API.
3. Students will practice databases, testing, security, and concurrency on real tasks.
4. Students will develop strong habits in communication, task management, and self-study.
5. Students will build a real project from scratch that can be used to showcase their skills on their CV.

COURSE POST REQUISITES: -

LITERATURE

- Let's Go, Alex Edwards.
- Let's Go Further, Alex Edwards.
- Go in Action, William Kennedy
- Effective Go.

COURSE ASSESSMENT CRITERIA

Assessment occurs continuously throughout the course. The evaluation will be based on the levels of (maximums in %):

Type of activity	Final scores
Participation	6%
Practice	24%
SIS	15%
Midterm/Endterm	15%
Final exam	40%
Total	100%

TASKS

for students' independent study (SIS)

Week	SIS	Cost (in points)
6	Checkpoint 1: Show running skeleton and database design	5
10	Checkpoint 2: Present one end to end slice work with auth	5
13	Checkpoint 3: Show dockerized application. Final project feedback will provided	5
	Total	15

TASKS
for practice

Week	TSIS	Cost (in points)
2	Initialize a Go module and run the app locally.	3
3	Add a router (Gin) and basic middleware.	3
4	Create an initial schema with migrations.	3
5	Build repository functions.	3
6	Add pagination, filtering, and sorting to one resource.	3
7	Implement a worker with goroutines and channels.	3
9	Add context-based cancellation and shutdown hooks.	3
12	Containerize the service with a Dockerfile.	3
	Total	24

COURSE CALENDAR

Week	Class work			SIS
	Topic	Lectures (hours)	Practice (hours)	SIS defense
1	**Go fundamentals:** • Tooling • Modules • Project layout	2	1	
2	**HTTP with net/http:** • Handlers • JSON • Middleware pattern • Routing intro	2	1	Practice 1. Go project setup
3	**Routers and configuration:** • Gin • Route groups • Middleware chains • Env/config • Dependency wiring	2	1	Practice 2. Routing and middleware
4	**Data modeling and migrations:** • Relational modeling • Keys and indexes • Migrations with migrate	2	1	Practice 3. Database migrations
5	**Docker:** • Mutli-stage builds • Compose • Env management • Best practices	2	1	Practice 8. Service Containerization

6	**database/sql and sqlx:** • Connection pooling • Queries and scans • Transactions • Context timeouts • Error handling	2	1	Practice 4. Repository implementation Checkpoint 1. Project foundation
7	**Query patterns and instrumentation:** • Pagination • Filtering • Sorting • N + 1	2	1	Practice 5. Add pagination, filtering, and sorting to one resource.
8	**Midterm**	2	1	
9	**Concurrency I:** • Goroutines • Channels • Worker pools	2	1	Practice 6. Background worker
10	**Concurrency II:** • Context propagation • Cancellation • Graceful shutdown • Timeouts	2	1	Checkpoint 2. Auth and one core flow
11	**Auth and security essentials:** • Password hashing • JWT • Sessions vs tokens • CORS • Input validation • Simple rate limiting	2	1	
12	**Testing strategy:** • Table-driven tests • httptest • Mocks via interfaces • Coverage	2	1	Practice 7. Graceful shutdown and timeouts
13	**Observability** • Structured logging • Metrics • Basic profiling	2	1	Checkpoint 3. Reliability and ops
14	**Performance and reliability** • Caching options • Idempotency • Retries with backoff • Circuit breaker • Rate limiting	2	1	
15	**Wrap-up and final preparation** **End-term**	2	1	Project defense
16-17	Final Exam			

Course assessment schedule

No	Assessment criteria	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	
1	Attendance / participation	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	6%
2	Practice		*	*	*	*	*	*		*			*				24%
3	SIS						*				*			*			15%
4	Midterm/ End-term								*							*	15%
6	Final exam																40%
	Total																100 %

Class sessions – will be a mixture of information, discussion and practical application of skills.
Practice – will reinforst the students knowledge by practical appliance of lectured materials.
In-class assessment – will prepare students for their mid-term and final assessment and identify the competence level they have achieved on a related subject matter, the aim being to diagnose potential discrepancies in students' understanding and performance in order to make specific adjustments to the course content and procedures and/or to assign additional assignments to certain individuals or the whole group.
Home assignments – will consolidate the concepts and materials taken during in-class activities, help students to expand the content through diverse background resources and/or practise certain skill areas; they will also develop the students' ability to work individually in exploring and examining related issues.
SIS (Student Independent Study) – comprises group Project to be done by students on the independent basis. Students are supposed to use knowledge and skills acquired in class to do the project. Assistance and advice will be provided by teachers during office hours.
TSIS (Teacher Supervised Student Independent Study) – student self-made project.
End-term test – a diagnostic test used to identify the students' progress, their strengths and weaknesses, intended to force student to prepare for Final Exam. It includes computer based test.
Final examination – 1) an attainment test designed to identify how successful the students have been achieving objectives.
Grading policy:
Intermediate attestations (on 8th and 15th week) join topics of all lectures, practice, laboratories, SIS, TSIS and materials for reading discussed to the time of attestation. he largest part of the course is dedicated to practical work, with a significant portion of the practice focused on team projects (SIS). This reflects the collaborative environment in real IT companies, where teamwork and proper communication hold equal importance with coding skills. The final project will be worked on throughout the entire course, with milestones and checkpoints to track progress and provide timely feedback, allowing students to stay on the right track.
Final exam is conducted in the form of final project defense. Maximum number of points is 40. At the end of the semester you receive overall total grade (summarized index of your work during semester) according to conventional KBTU grade scale.

ATTENTION!

1) If student missed more than **30% of lessons student receives «F (Fail)»** grade;
2) If for two attestations student receives 29.4 or less points, this student is not accepted to final exam and for all course he (she) receives **«F (Fail)» grade;**
3) If student receives 9.4 or less points on final exam, then independently on how many points he (she) received for two attestations, in whole he (she) receives **«F (Fail)» grade;**

In the case of missing or being late for final exam without plausible reason, independently on how many points he (she) received for two attestations, in whole he (she) receives **«F (Fail)» grade.**

ACADEMIC POLICY:

1. Cheating, duplication, falsification of data, plagiarism are not permitted under any circumstances!
2. Students must participate fully in every class. While attendance is crucial, merely being in class does not constitute "participation". Participation means reading the assigned materials, coming to class prepared to ask questions and engage in discussion.
3. Students are expected to take an active role in learning (the instructor will provide the information and guidelines to do this).
4. Students must come to class on time.
5. Students are to take responsibility for making up any work missed.
6. Make up tests in case of absence will not normally be allowed.
7. Mobile phones must always be switched off in class.
8. Students should always show tolerance, consideration and mutual support towards other students.

Grade		**Achievement percentage**	**Assessment criterion**
«Excellent»	A	95-100%	This grade is given when the student: demonstrated a complete understanding of the course material; did not make any errors or inaccuracies; completed control and laboratory work in a timely and correct manner, and submitted reports on them; demonstrated original thinking; submitted control quizzes on time and without any errors; completed homework assignments; engaged in research work; independently used additional scientific literature in studying the discipline; was able to independently systematise the course material.
	A -	90-94%	
«Good»	B+	85-89%	This grade is given when the student: Has mastered the course material at no less than 75%; Did not make gross errors in responses; Timely completed control and laboratory work and submitted them without fundamental remarks; Correctly completed and timely submitted control tests and homework assignments without fundamental remarks;
	B	80-84%	
	B-	75-79%	

	C+	70-74%	Utilized additional literature as indicated by the instructor; Engaged in research work, made non-fundamental errors and fundamental errors corrected by the student; Managed to systematise the course material with the help of the instructor.
«Satisfactory»	C	65-69%	This grade is given when the student: Has mastered the course material no less than 50%; Required assistance from the instructor when completing control and laboratory work, homework assignments; Made inaccuracies and non-fundamental errors when submitting control tests; Did not demonstrate activity in research work, relied solely on the educational literature indicated by the instructor; Experienced more difficulty in systematising the material.

Instructor **Salikh Pernebek**

Minutes #9 of School of Information Technology and Engineering meeting on January 5, 2026